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Y12

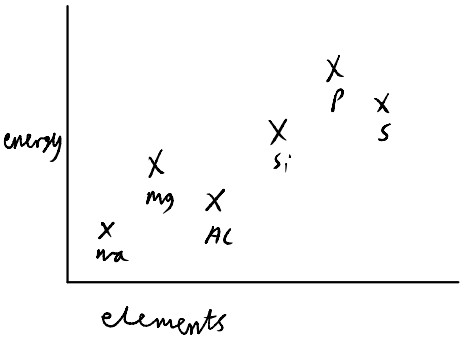
chemistry

Easter Revision Course AS (Y12)

1.

State and explain the general trend for the first ionisation energy across Period 3.
In addition, give one element which deviates from the general trend for the first ionisation energy across Period 3.
Explain why this deviation occurs in detail. (7 Marks)

Ionisation energy increases across the period generally. As proton number increases/nuclear charge. Shielding stays the same as one e^- added to same shell across the period.
So stronger electrostatic attraction of valence e^- to the nucleus.



Al lower IE than Mg because in Al we popvlate the 3p orbital which is higher in energy and further away from the nucleus so less energy needed to release 3p e^-.

S lower than P as e^- pair up in 3p orbital so e^-/e^- repulsion means it is easier to remove one e^-.

2.

This sample of krypton has four peaks on the mass spectrum.
The table below shows the data for the spectrum. Hence, determine the relative atomic mass (A_r) of this sample of krypton. Provide your answer in 1 decimal place.

m/z	82	83	84	86
Relative Intensity	6	1	28	8

$$\frac{(82 \times 6) + (83 \times 1) + (84 \times 28) + (86 \times 8)}{43} = \frac{3615}{43} = 84.1$$

$$\frac{\sum(m/z \times \text{relative})}{\sum \text{relative intensity}}$$

3.

A sample of hydrated nickel sulphate ($NiSO_4.xH_2O$) with a mass of 2.287 g was heated to remove all water of crystallisation, leaving the solid with a mass of 1.344 g. Suggest how the student can ensure that all the water has been removed and calculate the value of x.
(6 Marks)

$NiSO_4 \cdot xH_2O$ $\quad n = \frac{2.287}{58.7+32.1+16\times4+18x}$

$NiSO_4 \xrightarrow{\Delta} NiSO_4 + xH_2O$

$2.287g \quad 1.344g \quad 0.943$

$n = \frac{0.943}{18} = 0.0524$

$2.287 - 1.344 = 0.943$ $\quad n(NiSO_4) = \frac{1.344}{154.4} = 8.7\times10^{-3}$

$\frac{0.0524}{8.7\times10^{-3}} = 6 \quad \therefore x = 6$

Heat the solid until the mass remains constant.

4.

If the blood alcohol level of a driver is over 80 mg of ethanol per 100 cm^3 of blood, an offence is committed.
What is the concentration, in mol dm^{-3}, of ethanol if there are 80 mg of ethanol (M_r = 46.0) per 100 cm^3 of blood?
(1 Mark)

a) 0.00017
b) 0.0017
c) 0.017
d) 1.7

80 mg = 0.08 g

$n = \frac{0.08}{46} = 1.7 \times 10^{-3}$

Conc = $\frac{mol}{vol}$

$\frac{1.7 \times 10^{-3}}{0.1} = 1.7 \times 10^{-2}$

100 cm^3 = 0.1 dm^3

5.

Potassium nitrate, KNO_3, decomposes on strong heating, forming oxygen and solid **A** as the only product.
A 1.00 g of KNO_3 was heated strongly until fully decomposed into **A.**
Given that potassium nitrate has M_r = 101.1
Oxygen gas produced occupied a volume of 1.22×10^{-4} m^3 after decomposition at 298 K and 100 kPa.
Calculate the number of moles of KNO_3 in the 1.00 g sample. State the ideal gas equation and use it to calculate the number of moles of oxygen produced in this decomposition. (5 Marks)

$kNO_3 \xrightarrow{\Delta} A + O_2$

1.00g

101.1

$pV = nRT$

P = Pa
V = m^3
n = moles
R = 8.31 $j\,mol^{-1}\,k^{-1}$
T = kelvin

P = 100,000 Pa
V = 1.22×10^{-4} m^3
n = N
R = 8.314 $j\,mol^{-1}\,k^{-1}$
T = 298k

$N = \frac{pV}{RT} = \frac{100{,}000 \times 1.22 \times 10^{-4}}{8.31 \times 298} = 4.93 \times 10^{-3}$ mol O_2

$V \propto T$ Charles' law

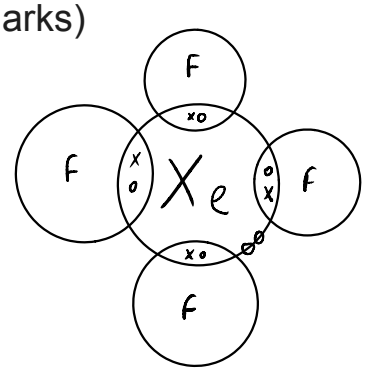
$V \propto N$ Avagadoros

$V \propto \frac{1}{P}$ Boyle's Law

$n(kNo_3) = \frac{1}{101.1} = 9.89 \times 10^{-3}$

6.

The melting point of XeF_4 is higher than the melting point of PF_3
Explain why these two compounds have different melting points.
In your explanation, you should include the shape of each molecule and how the shape influences the forces that affect the melting point.
(6 Marks)

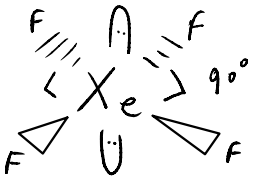


Xe = group 8 ∴ 8 valence e⁻

4 x F = 4

12 e⁻ ⇒ 6 e⁻ pairs

4 bonded pairs (F)
2 lone pairs



90°

Square Planar

London Forces

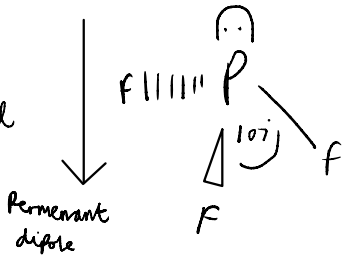
PF_3

P ⇒ 5 Ve⁻

3F ⇒ 3e⁻

8 e⁻ ⇒ 4 e⁻ pairs

3 Bp, 1 lp } Pyramidal



109.5 − 2.5 = 107

London forces
dipole - dipole forces

XeF_4 has stronger intermolecular forces as stronger london forces ⇒ XeF_4 has more e⁻ than PF_3.

more e⁻ means stronger london forces.
larger surface area increases london forces.

F is more electronegative.

The pairs arrange to maximise seperation and maximise repulsion.

lp / lp repulsion > lp/bp > bp/bp

7.

Which molecule has the largest dipole?
(1Mark)

a) ClF_3
b) BF_3
c) SF_6
d) CF_4

8.

The enthalpy change when ethanoic acid reacts with sodium hydroxide is,
$\Delta H = -56.1$ kJ mol^{-1}

$$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$$

The equation for the reaction is shown above.
Calculate the temperature rise when 25 cm^3 of 2.0 mol dm^{-3} aqueous ethanoic acid react with 25 cm^3 of 2.0 mol dm^{-3} aqueous sodium hydroxide.
Assume that both the solution has density of 1.0 g cm^{-3} and a specific heat capacity of 4.18 J K^{-1} g^{-1}
(4 Marks)

9.

An experiment on a pure sample of 2-methylpropan-2-ol was carried out to determine its enthalpy of combustion.
A sample of the alcohol was placed into a spirit burner and positioned under a beaker containing 50 cm^3 of water.
The spirit lamp was lit for a few minutes before extinguished.
The results are shown below:

Initial temperature of the water / °C	18.1
Final temperature of the water / °C	45.4
Initial mass of spirit burner and alcohol / g	208.80
Final mass of spirit burner and alcohol / g	208.58

Calculate a value for the heat energy released from the combustion of 2-methylpropan-2-ol.
Given that the specific heat capacity of water is 4.18 J K^{-1} g^{-}

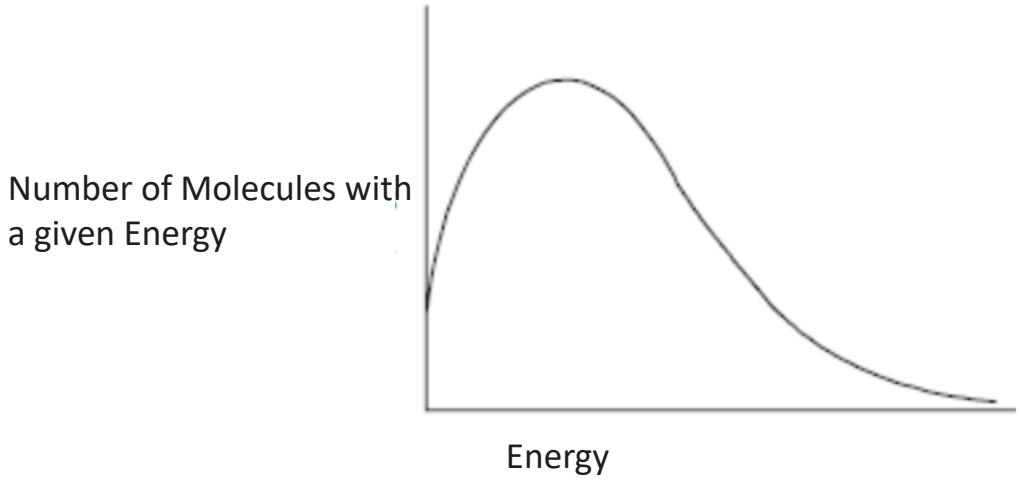
In addition, calculate the amount, in moles, of 2-methylpropan-2-ol burned in the experiment.
Hence, calculate the enthalpy of combustion of 2-methylpropan-2-ol, in kJ mol^{-1}
(5 Marks)

10.

On a labelled axis, draw a curve to represent the Maxwell–Boltzmann distribution of molecular energies in a gas. Label this curve $\mathbf{T}_1$. and on the same axis, draw a second curve to represent the same sample of gas at a lower temperature. Label this curve $\mathbf{T}_2$. Explain why a small decrease in temperature can lead to a large decrease in the rate of a reaction.

(8 Marks)

11.



What does the area under the graph represent? (1 Mark)

a) total energy.
b) activation energy.
c) total number of reacting molecules.
d) total number of molecules present.

12.

A pale brown mixture of NO_2 and N_2O_4 is allowed to reach equilibrium in a sealed gas syringe.
The equation for the mixture is as follows:

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

When the pressure increases within the syringe and the mixture becomes paler in colour.
When the syringe is placed in a hot oven the mixture becomes darker in colour.
Which of the following statements is correct? (1 Mark)

a) NO_2 is brown and the forward reaction is exothermic.
b) NO2 is brown and the forward reaction is endothermic.
c) NO2 is colourless and the forward reaction is exothermic.
d) NO2 is colourless and the forward reaction is endothermic.

13.

Below shows the equation for the process of manufacturing hydrogen from methane.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \qquad \Delta H = +206\ kJ.mol^{-1}$$

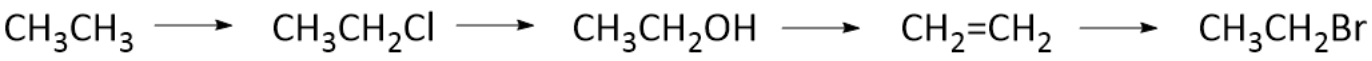
Using Le Chatelier's principle state whether a high or low temperature should be used to obtain the highest possible equilibrium yield of hydrogen from this first stage.
Explain your answer.
In addition, A nickel catalyst is used.
Explain why the catalyst is more effective when coated onto an unreactive honeycomb.
(5 Marks)

14.

Chlorofluorocarbons (CFCs) are a group of halogenoalkanes currently banned in various countries this is because of their effect on the environment. Hence, identify the **radical** produced from this CFC that is responsible for the depletion of ozone in the atmosphere and explain with the aid of equations, why a single radical can cause the decomposition of many molecules of ozone.
(4 Marks)

15.

Which one of the following mechanisms is **not** involved in the reaction sequence below? (1 Mark)



a) electrophilic addition
b) electrophilic substitution
c) nucleophilic substitution
d) free-radical substitution

16.

Alkenes react with bromine (Br_2). Provide the name and outline the mechanism for the reaction of cyclohexene with Br_2.
In addition, explain why there is an attraction between a C=C double bond and Br_2
(8 Marks)

17.

When 2-bromobutane is warmed with potassium hydroxide solution, both substitution and elimination reactions occur. What is the role of the hydroxide ions in the elimination reaction? (1 Mark)

a) Base
b) Catalyst
c) Nucleophile
d) Electrophile

18.

A chlorofluoroalkane, **A**, contains by mass 51.6% fluorine, 32.1% chlorine and no hydrogen.

Chlorine exists as two isotopes, ^{35}Cl and $3^{37}Cl$, in the ratio 3:1

Fluorine only exists as one isotope, ^{19}F.

A mass spectrum of **A** is obtained using electron impact ionisation. The mass spectrum shows three molecular ion peaks at *m/z* = 220, 222 and 224. Hence, determine the formula of each of the three molecular ions of **A**. Predict and explain the ratio of the relative abundancies of each of the three molecular ion peaks.

(6 Marks)

19.

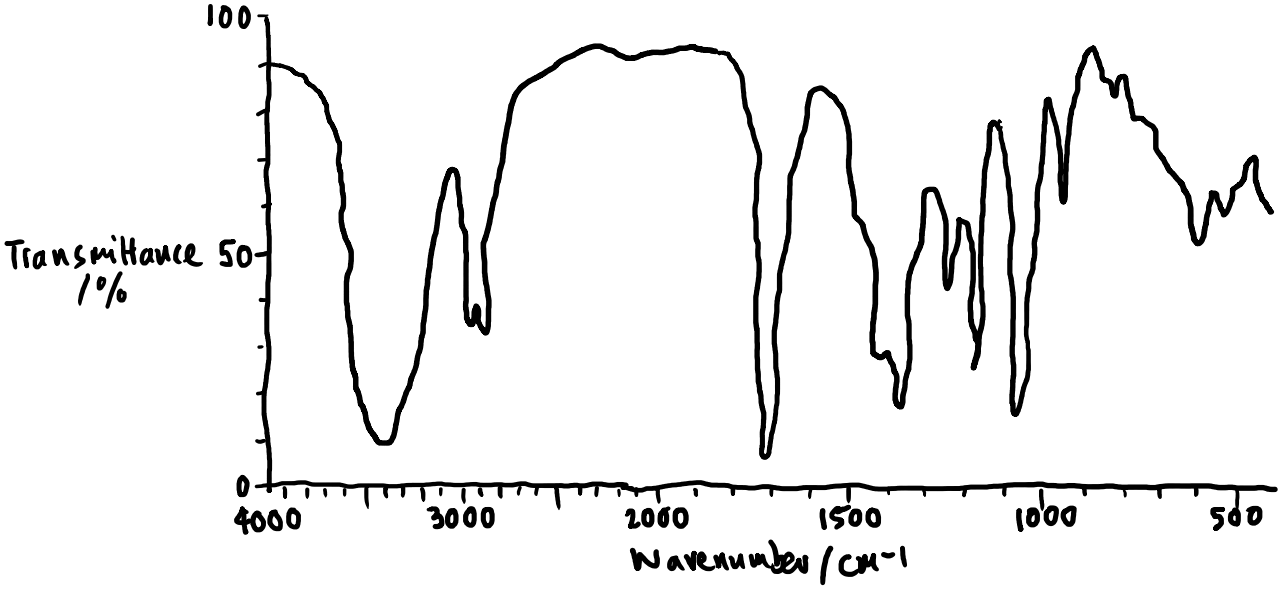
For a sample of 2-methylpropan-2-ol and butan-2-ol, identify a reagent or combination of reagents that the could be used to distinguish between these alcohols. State what would be observed for each alcohol.
(3 Marks)

20.

There are four isomeric alcohols with the molecular formula $C_4H_{10}O$
Two of these are butan-1-ol ($CH_3CH_2CH_2CH_2OH$) and butan-2-ol.
The other two isomers are alcohol **X** and alcohol **Y**.
Draw the displayed formula for butan-2-ol and give the structure for alcohol **X**
Alcohol X does not react with acidified potassium dichromate(VI) solution.
Give the name of the fourth isomer.
(3 Marks)

21.

The infrared spectrum of an organic compound is shown.

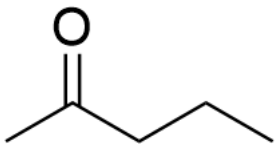


Which compound produces this spectrum? (1 Mark)

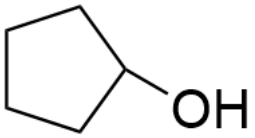
a) ethanoic acid
b) 4-hydroxybutanone
c) propan-1-ol
d) prop-2-en-1-ol

22.

Below are compounds A and B, shown below, are isomers of $C_5H_{10}O$



A



B

Deduce the number of peaks in the proton n.m.r. spectrum of **A**
In addition, identify a reagent that you could use to distinguish between **A** and **B**.
State what would be observed when the compound is treated with this reagent.
(4 Marks)

23.

Deduce which of Na^+ and Mg^{2+} is the smaller ion.
Explain your answer.
(2 Marks)

24.

Identify the Period 3 element that has the highest melting point.

Explain your answer based on the structure and bonding.

(4 Marks)

25.

Below shows the equation for Haber process

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Discuss the role of iron is used as a heterogeneous catalyst in the process.

Include these in your answer.

- The meaning of the term heterogeneous catalyst
- How iron acts as a heterogeneous catalyst
- The factors that affect the efficiency and lifetime of the catalyst.

(6 Marks)

26.

A sample of the metal silver has the relative atomic mass of 107.9 and there are two isotopes in existence.

There are 54.0% of the silver atoms are one isotope with a relative mass of 107.1 in this sample.

Calculate the relative mass of the other silver isotope.

Explain why the isotopes of silver have identical chemical properties.

(4 Marks)

27.

Draw the shape of BrF_3 and predict its bond angle.
Include any lone pairs of electrons that influence the shape.

Shape of BrF_3

In addition, draw the shape of BrF_4^- and predict its bond angle.
Include any lone pairs of electrons that influence the shape.

Shape of BrF_4^-

(4 Marks)

28.

The forward reaction in this equilibrium is endothermic

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

Which of the statement below are correct?

a) If the total pressure is increased at constant temperature, the proportion of $COCl_2$ in the equilibrium mixture will decrease
b) Use of a catalyst will increase the proportion of $COCl_2$ in the equilibrium mixture at constant temperature and pressure
c) Reducing the equilibrium concentration of CO will increase the value of the equilibrium constant
d) Raising the temperature from 373 K to 473 K will increase the value of the equilibrium constant

29.

Which of the following statement regarding the trend in atomic radius across Period 3 of the Periodic Table are correct?

a) radius increases because the atoms have more electrons
b) radius decreases because nuclear charge increases
c) radius increases because shielding (screening) increases
d) radius decreases because shielding (screening) decreases

(1 Mark)

30.

Chlorine has a low boiling point because the forces between the molecules are weak.

Explain how these forces arise between molecules of chlorine.

31.

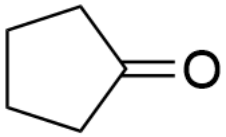
Define the term electronegativity.
Explain the polarity of the C–Br bond in 2-bromopropane.
(3 Marks)

Electronegativity:
Explanation:

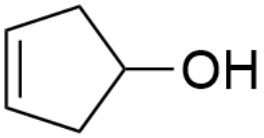
32.

Isomers **A** and **B** have the molecular formula C_5H_8O

Isomer A



Isomer B



Explain how and why isomers **A** and **B** can be distinguished by comparing **each** of their

- boiling points
- ^{13}C NMR spectra
- infrared spectra.

(6 Marks)

33.

Which compound does **not** show stereoisomerism? (1 Mark)

a) 1,2-dichloropropene
b) 1,2-dichloropropane
c) 1,3-dichloropropene
d) 1,3-dichloropropane

34.

The table shows some standard enthalpies of formation.

	$CO_2(g)$	$H_2(g)$	$CH_3OH(g)$	$H_2O(g)$
$\Delta H_f^{\ominus}$/kJ mol^{-1}	– 394	0	– 201	– 242

Use these standard enthalpies of formation to calculate a value for the standard enthalpy change of this synthesis

$$CO_2(g) + 3H_2(g) \rightleftharpoons CH_3OH(g) + H_2O(g)$$

(3 Marks)

35.

Write an equation for the incomplete combustion of C_8H_{18} to form carbon monoxide and water only.

A catalytic converter is used to remove carbon monoxide from the exhausts of cars. Identify a catalyst used in the catalytic converter.

Write an equation to show how carbon monoxide is removed in a catalytic converter. In addition, state why the water produced in the exhaust gases may contribute to global warming.

(4 Marks)

36.

The petrol fraction obtained from crude oil can be used as fuel in cars.

State the meaning of fraction, as used in the term petrol fraction.

In addition, hexadecane ($C_{16}H_{34}$) can be cracked at high temperature to form petrol.

Complete the equation below and give the name of a catalyst used in this cracking reaction.

(4 Marks)

37.

Which compound can be oxidised to form $(CH_3)_2CHCOCH_3$? (1 Mark)

a) 2-methylpropan-1-ol
b) 2,2-dimethylpropanol
c) 2-methylbutan-2-ol
d) 3-methylbutan-2-ol

38.

Ethanol can be oxidised by acidified potassium dichromate(VI) to ethanoic acid in a two-step process.
The process is shown below:

ethanol → ethanal → ethanoic acid

In order to ensure complete oxidation, the reaction is carried out under reflux.
Describe what happens when a reaction mixture is refluxed and why it is necessary.
(3 Marks)

39.

A 5.0 g of an oxide contains 4.0 g of molybdenum.
Hence, what is the empirical formula for this oxide. (1 Mark)

a) MoO_2
b) MoO_5
c) Mo_2O_3
d) Mo_3O_2

40.

Which shows the major product(s) formed when chlorine reacts with cold, dilute, aqueous sodium hydroxide?

a) $NaCl$ only
b) $NaClO$ only
c) $NaCl$ and $NaClO$
d) $NaCl$ and $NaClO_3$

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